Project Suggestion 6 — Profit-Maximizing Model for Prediction Markets Under Crowd (Majority) Betting

Motivation (Why this topic is important)

Prediction markets (e.g., Kalshi-style event contracts) aggregate beliefs into prices, but prices are not always "perfect probabilities." Market prices can reflect **behavioral biases**, **herding**, and **liquidity effects**, especially when a large share of participants follow similar heuristics or "Bet-AI" style signals.

This creates an applied economics question with real industry relevance:

- **How do you build a trading / quoting strategy that maximizes expected profit under realistic market microstructure and biased crowds?**
- **When does the majority become systematically wrong, and how can a model exploit that while controlling risk?**

This project is useful because it connects:

- **Behavioral economics** (biases like favorite–longshot bias, overreaction, herding)
- **Market microstructure** (bid/ask spreads, order books, fees, liquidity)
- **Decision theory** (optimal bet sizing, utility, drawdown control)
- **ML** (forecasting true probabilities, detecting bias regimes)

A good outcome is a "playbook" for designing safer, more robust decision policies for prediction-market trading—without assuming the market is perfectly efficient.

Type of project

Applied (empirical + simulation).
Goal: build a **profit-maximizing policy** in a **simulated Kalshi-like market**, using public historical data (when available) or synthetic market data calibrated to known stylized facts.

Note: This should be framed as **research + simulation**, not "guaranteed money." Include responsible gambling / risk controls.

Model framing

A. Economic baseline: rational expected value + risk control

1 1 Model each contract outcome as Bernoulli with true probability p.

2 2 Market price q (contract cost) approximates an implied probability, but can be biased.

3 3 You enter trades when you believe $p-q$ (your edge) exceeds costs/fees/spread.

B. Behavioral overlay: “majority bettor” bias and regime detection

In many betting markets, systematic patterns appear, like the **favorite–longshot bias** (prices/odds biased by preferences for longshots) and other risk-preference distortions. You can model this as:

- • **Crowd bias function**: $q=f(p,\text{sentiment},\text{liquidity},\text{attention})$
- • Identify regimes where the crowd overprices “exciting” tail outcomes.

Behavioral references on bias are well established in gambling markets literature.

C. Profit objective: maximize expected log-growth (or utility)

Use **Kelly-style** optimal sizing (or fractional Kelly) to control risk while optimizing long-run growth.

What the student builds

1) A probability model (“true p” estimator)

Choose one target domain, e.g.:

- • Elections/politics (harder; needs careful sourcing)
- • Economic indicators
- • Sports (if available data)
- • Weather-style events
- • Any binary outcomes with public signals

Approaches:

- Logistic regression / gradient boosting on engineered features
- Bayesian model (posterior over p)
- Ensemble probability calibration (Platt scaling / isotonic)

2) A microstructure-aware trading policy

Implement a policy that decides:

- **Enter / exit** based on edge $p-q$
- **Size** based on Kelly or constrained utility (max drawdown, VaR)
- **Avoid** trading when spreads/fees erase edge

Kalshi-like contracts resolve to $1 or $0, and markets rely on liquidity providers / spreads. (High-level mechanics described in mainstream coverage.)

3) A "majority bettor / Bet-AI crowd" bias detector

Design indicators like:

- Price momentum + volume spikes (herding proxy)
- Mispricing persistence (price stays away from your p)
- Longshot overpricing proxies (tail outcomes)

Then condition the policy:

- Normal regime: trade small edges
- Herding regime: trade only high-conviction edges (or fade the crowd)

4) Backtesting and simulation

Deliver a reproducible backtest:

- P&L distribution
- Sharpe-like metrics (careful: not stock returns, but still useful)
- Max drawdown
- Hit rate and calibration (Brier score)

(Scoring rules like the **Brier score** are standard for probability accuracy evaluation.)

Deliverables

Research report (10–20 pages)

- Market setup and assumptions
- Behavioral hypotheses (majority bias)
- Model for p (true probability)
- Trading policy and risk controls
- Backtest results + limitations
- Ethics / responsible gambling section

Code repository

- Data ingestion / cleaning
- Probability model training + calibration
- Trading simulator (fees, spreads, slippage assumptions)
- Strategy evaluation notebooks

Experiment dashboard

- Calibration curves
- P&L curves
- Regime analysis plots (crowd-bias detector)

Final presentation

  - Clear summary of when the crowd is biased
  - What strategy does in each regime
  - Quantitative results and failure cases

Suggested references

Prediction markets & crowd aggregation

- Dai (2021), *The wisdom of the crowd and prediction markets* (discusses scoring rules and aggregation issues).
- Related working paper discussion: majority rule vs market aggregation inconsistencies (helpful for the "majority bettor" angle).

Behavioral economics in betting markets

- Snowberg & Wolfers (2010), NBER working paper on favorite–longshot bias (classic, highly cited).
- Berkowitz (2017), empirical favorite–longshot bias evidence.

Optimal bet sizing / decision theory

- Kelly criterion overview and background (start here; then students should cite the original 1956 paper in their report).
- Practical discussion and common misunderstandings (good to refine assumptions like utility, finite-horizon constraints).

Kalshi / microstructure intuition (background)

- High-level overview of how prediction-market trading and market makers work in practice (context for spreads/liquidity risk).
- https://www.federalreserve.gov/econres/feds/kalshi-and-the-rise-of-macro-markets.htm Links to an external site.